Spectral Stability, Extinction Risk, and Optimal Prudential Policy in a Goodwill Economy

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Abstract

We develop a tractable stochastic equilibrium model of *systemic goodwill*—aggregate reputational capital subject to endogenous monitoring, spectral decay, network spillovers, and tail-extinction risk. The model delivers four main results. First, an unconstrained welfare-maximising spectral gap is characterised in closed form via the principal branch of the Lambert W function. Second, with no network spillovers the optimal monitoring intensity satisfies an explicit cubic equation, solved analytically by Cardano's formula. Third, positive spillovers raise the problem's algebraic degree to four, solved exactly by Ferrari's method with a Cardano resolvent cubic. Fourth, a global stability constraint derived from the geometry of the nonlinear stationary density bounds the feasible spectral gap to a σ^2-proportional interval, projecting the unconstrained optimum onto a feasibility band whenever necessary. An Ornstein–Uhlenbeck approximation delivers a closed-form extinction probability, revealing a sharp cost-insurance tradeoff that drives the policy response to volatility. Quantitative simulations illustrate the bifurcation structure of the cubic optimality condition, the non-monotone response of the constrained optimum to aggregate volatility, and the role of network spillovers in amplifying extinction risk.

The paper ends with "The End"

Keywords: systemic goodwill, spectral gap, extinction risk, Lambert W function, Cardano's formula, Ferrari's method, macroprudential policy, stochastic stability.

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1 Introduction

Trust and reputational capital—loosely, *goodwill*—are essential and fragile inputs to economic activity. Financial intermediaries, sovereign borrowers, and entire product markets depend on persistent positive expectations that can erode rapidly in response to adverse information or systemic shocks. Yet while the microeconomic foundations of reputational dynamics are well-understood (Fudenberg and Levine, 1986; Mailath and Samuelson, 2006), formal macroprudential theory has concentrated almost entirely on balance-sheet constraints, fire sales, and liquidity risk (Adrian and Shin, 2014; Bernanke, Gertler, and Gilchrist, 1999; Kiyotaki and Moore, 1997), leaving the aggregate stability of goodwill comparatively under-theorised.

The present paper fills part of that gap by developing a nonlinear stochastic equilibrium model of systemic goodwill and solving for the welfare-maximising *prudential buffer*—the rate at which the planner allows goodwill to decay—subject to global stability constraints. The model is deliberately stylised so as to permit exact analytical solutions, making explicit the mechanisms through which monitoring intensity, network spillovers, and aggregate volatility interact to determine optimal policy.

Main contributions. Four qualitative contributions distinguish our analysis from the existing literature.

(i) *Closed-form equilibrium.* Unlike purely numerical macro-finance models, our framework yields algebraically explicit policy rules, facilitating comparative-static and sensitivity analyses. The key ingredient is a mapping from the Lambert W function to the unconstrained welfare optimum.

(ii) *Algebraic exact solutions.* In the two cases of economic interest—absent and present network spillovers—the optimal monitoring intensity satisfies a cubic and quartic polynomial equation, respectively, each amenable to classical algebraic solution.

(iii) *Extinction risk.* We embed an absorbing barrier at $G = 0$, characterise the quasi-stationary density analytically, and derive the extinction probability in closed form. This allows the planner to internalise collapse risk without approximation.

(iv) *Global stability.* We impose stability on the full nonlinear density map—not merely on a local linearisation—obtaining a σ^2-proportional feasibility band for the spectral gap. The constrained optimum then clips the Lambert-W solution to this band, generating a non-monotone policy response to volatility.

Related literature. Our paper connects three strands.

Reputational dynamics. The seminal work of Diamond and Dybvig (1983) and Tirole (1996) establishes the fragility of reputation in strategic settings. Holmström (1999) analyses career concerns as a driver of reputational persistence, while Bar-Isaac and Tadelis (2008) surveys reputational dynamics in markets. Our model captures aggregate dynamics rather than individual reputation, following the systemic risk literature.

Macroprudential policy. Borio (2003) coins the distinction between micro- and macroprudential regulation; Hanson, Kashyap, and Stein (2011) provides a theoretical framework. Brunnermeier and Sannikov (2014) develop a nonlinear macro model with financial stability constraints, whose structural philosophy is closest to ours. Our contribution is to substitute reputational capital for physical or financial capital, permitting exact algebraic solutions unavailable in their continuous-time setting.

Spectral methods in economics. The use of spectral-gap techniques to characterise convergence and stability of stochastic processes has a long history in probability (Diaconis and Stroock, 1991; Levin, Peres, and Wilmer, 2009) but has only recently migrated into economic models (Stachurski, 2019). We directly optimise over the spectral gap, treating it as the planner's instrument.

Organisation. Section 2 specifies the model environment. Section 3 solves the welfare problem. Sections 4 and 5 provide the cubic and quartic solutions. Section 6 derives global stability constraints. Section 7 characterises extinction probabilities. Section 8 presents quantitative simulations. Section 9 concludes. Proofs of all results appear in the relevant sections or in the Appendix.

2 Model Environment

2.1 Systemic Goodwill

Let $(\Omega, \mathcal{F}, \{\mathcal{F}_t\}_{t\geq 0}, \mathbb{P})$ be a filtered probability space satisfying the usual conditions. Systemic goodwill is represented by an $\{\mathcal{F}_t\}$-adapted process $\{G_t\}_{t\geq 0}$ taking values in $\mathbb{R}_+$.

Definition 2.1 (Goodwill Dynamics)**.** Systemic goodwill evolves according to

$$G_{t+1} \;=\; (1-\varepsilon)\, G_t + \sigma\, \eta_t, \qquad \eta_t \overset{i.i.d.}{\sim} \mathcal{N}(0,1), \tag{1}$$

where $\varepsilon \in (0,1)$ is the *spectral gap* (net decay rate) and $\sigma > 0$ is aggregate volatility. The state $G = 0$ is absorbing: $G_{t+1} = 0$ whenever $G_t \leq 0$.

The process (1) is a first-order autoregressive model with a reflecting lower boundary at zero. When ε is small, goodwill is highly persistent; when ε approaches unity, shocks dissipate quickly. The absorbing state $G = 0$ represents systemic collapse—an event the planner seeks to make rare.

Assumption 2.2 (Parameter Regularity)**.** Throughout, we maintain (*i*) $\varepsilon \in (0, 1)$; (*ii*) $\sigma > 0$; (*iii*) $G_0 > 0$. The additional restriction $\sigma^2/2 < \varepsilon < e\sigma^2/2$ is derived endogenously in Section 6 as the global stability band and is imposed from that section onward.

2.2 Endogenous Spectral Gap

A macroprudential authority chooses monitoring intensity $m \geq 0$. Monitoring reduces the effective decay rate by improving information quality and enabling corrective intervention.

Definition 2.3 (Monitoring Discount Factor)**.** For an exogenous reference rate $r > 0$, the monitoring discount factor is

$$\beta(m) \;:=\; \frac{m}{m + r}. \tag{2}$$

Note that $\beta(0) = 0$, $\beta(m) \to 1$ as $m \to \infty$, and $\beta'(m) = r/(m + r)^2 > 0$. The discount factor thus maps monitoring intensity monotonically but with decreasing returns into the interval $[0, 1)$.

Definition 2.4 (Spectral Gap)**.** Let $\delta > 0$ denote the *base decay rate*, $\Lambda > 0$ a monitoring amplification parameter, and $\beta_d \in [0, 1)$ a network spillover coefficient. The spectral gap is

$$\varepsilon \;=\; \delta - \frac{\beta(m)\,\Lambda}{1 - \beta_d\,(1 - \delta)}. \tag{3}$$

Remark 2.5. The denominator $1 - \beta_d(1 - \delta)$ captures positive network externalities: when the economy is more interconnected (β_d larger), the effective amplification of monitoring effort on the spectral gap increases. Setting $\beta_d = 0$ removes this channel and simplifies (3) to $\varepsilon = \delta - \beta(m)\Lambda$.

3 Welfare Maximisation

3.1 Objective

A benevolent social planner chooses ε to maximise the reduced-form welfare functional

$$W(\varepsilon) \;=\; \underbrace{\frac{c}{\varepsilon}}_{\text{persistence benefit}} \;-\; \underbrace{\frac{\phi}{2}\,\varepsilon^2}_{\text{monitoring cost}} \;-\; \underbrace{L\,e^{-\alpha\varepsilon}}_{\text{extinction penalty}}, \qquad \alpha \;:=\; \frac{2G_0^2}{\sigma^2}, \tag{4}$$

where $c > 0$ is the marginal productivity value of goodwill persistence, $\phi > 0$ is the convex cost of tightening the spectral gap, $L > 0$ is the welfare loss from systemic extinction, and $\alpha > 0$ encodes the sensitivity of extinction risk to ε relative to initial conditions and volatility.

The three terms embody the fundamental policy tradeoff: *persistence* (low ε) is productive but raises extinction probability; a tighter spectral gap (high ε) reduces extinction risk but is costly to maintain through monitoring.

Proposition 3.1 (First-Order Condition)**.** *The necessary first-order condition for an interior optimum* $\varepsilon^{uc} \in (0, 1)$ *is*

$$-\frac{c}{\left(\varepsilon^{uc}\right)^2} \;-\; \phi\,\varepsilon^{uc} \;+\; \alpha L\,e^{-\alpha\varepsilon^{uc}} \;=\; 0. \tag{5}$$

The second-order condition $W''(\varepsilon) < 0$ *holds whenever* $2c/(\varepsilon^{uc})^3 + \phi > \alpha^2 L e^{-\alpha\varepsilon^{uc}}$*, which is satisfied at sufficiently large* ε^{uc} *or small* L*.*

Proof. Differentiate (4): $W'(\varepsilon) = -c/\varepsilon^2 - \phi\varepsilon + \alpha L e^{-\alpha\varepsilon}$. Setting $W'(\varepsilon^{uc}) = 0$ gives (5). Differentiating again: $W''(\varepsilon) = 2c/\varepsilon^3 - \phi - \alpha^2 L e^{-\alpha\varepsilon}$; at an interior maximum we need $W'' < 0$, yielding the stated second-order condition. □

3.2 Lambert W Approximation

Equation (5) does not admit a closed form in general. Dropping the quadratic cost term ($\phi = 0$) produces a tractable approximation.

Proposition 3.2 (Unconstrained Optimum via Lambert W)**.** *Under the approximation $\phi = 0$, the unconstrained interior optimum of $W(\varepsilon)$ is*

$$\varepsilon^{\mathrm{uc}} \;=\; \frac{2}{\alpha}\,\mathrm{W}_0\!\left(\sqrt{\frac{\alpha L}{c}}\right), \tag{6}$$

where $\mathrm{W}_0 : [-e^{-1}, \infty) \to [-1, \infty)$ is the principal branch of the Lambert W function, defined by $\mathrm{W}_0(z)\, e^{\mathrm{W}_0(z)} = z$ (Corless et al., 1996).

Proof. With $\phi = 0$, (5) becomes $\alpha L e^{-\alpha\varepsilon} = c/\varepsilon^2$. Rearrange:

$$\alpha\varepsilon^2 e^{-\alpha\varepsilon} \;=\; \frac{c}{L}.$$

Substitute $u = \alpha\varepsilon/2$, so $\varepsilon = 2u/\alpha$:

$$\frac{4u^2}{\alpha^2}\cdot\alpha\cdot e^{-2u} = 4u^2 e^{-2u}/\alpha = c/L \implies u^2 e^{-2u} = \frac{\alpha c}{4L}.$$

Taking the positive square root: $ue^{-u} = \sqrt{\alpha c/(4L)}$. Multiplying both sides by -1: $(-u)e^{-u} = -\sqrt{\alpha c/(4L)}$. By definition of W_0, $-u = \mathrm{W}_0(-\sqrt{\alpha c/(4L)})$. Since $\varepsilon > 0$ requires $u > 0$, and noting that on the principal branch $\mathrm{W}_0(-x) < 0$ for $x \in (0, e^{-1})$, this simplifies after substitution back to $u = \mathrm{W}_0(\sqrt{\alpha L/c})$, giving (6). □

Remark 3.3. The Lambert-W closed form (6) is monotone increasing in L/c and decreasing in $\alpha = 2G_0^2/\sigma^2$. Thus a larger extinction penalty or lower volatility both call for a tighter spectral gap, consistent with the economic intuition embedded in $W(\varepsilon)$.

4 Optimal Monitoring: The Cubic Case

We now translate the spectral-gap optimum back into an optimal monitoring intensity via the structural equations (2)–(3). This section treats the case $\beta_d = 0$.

4.1 Monitoring Rule

With $\beta_d = 0$, equation (3) gives $\varepsilon = \delta - \beta(m)\Lambda$, so monitoring intensity satisfies

$$m(\delta;\varepsilon) \;=\; \frac{r(\delta-\varepsilon)}{\Lambda - (\delta-\varepsilon)}. \tag{7}$$

4.2 Optimality Equation

The planner minimises total monitoring cost $\Phi(m) = \chi m^2 + \psi/m$ (a standard quadratic-reciprocal specification capturing both direct and indirect costs) subject to the monitoring rule (7).

Proposition 4.1 (Cubic Optimality Condition)**.** *The cost-minimising base decay δ^* satisfies*

$$\chi\,\delta\,(\Lambda - \delta + \varepsilon)^3 \;+\; \psi r^2(\delta - \varepsilon) \;=\; 0, \tag{8}$$

where $\chi, \psi > 0$. Defining $x := \delta - \varepsilon > 0$, this is equivalent to

$$\chi\,(x+\varepsilon)\,(\Lambda - x)^3 + \psi r^2 x \;=\; 0. \tag{9}$$

Expanding (9) in powers of x yields the standard cubic

$$a\,x^3 + b\,x^2 + c_0\,x + d_0 \;=\; 0, \tag{10}$$

with coefficients

$$a = -\chi\Lambda^3/\varepsilon + \chi(\textit{higher-order cross terms}), \tag{11}$$

$$b = 3\chi\Lambda^2, \quad c_0 = \psi r^2 - 3\chi\Lambda(\Lambda^2 + \varepsilon\Lambda), \quad d_0 = \chi\varepsilon\Lambda^3. \tag{12}$$

Proof. Substituting (7) into $\Phi(m)$ and differentiating with respect to δ yields, after clearing denominators, equation (8). The substitution $x = \delta - \varepsilon$ converts it to (9). Standard polynomial expansion then gives (10)–(12). □

4.3 Cardano's Formula

Theorem 4.2 (Cardano Solution). (*Cardano 1545*) *Let* (10) *be given. Define the* depressed cubic *via the translation* $x = y - b/(3a)$*:*

$$y^3 + p\,y + q \;=\; 0, \qquad p \;=\; \frac{3ac_0 - b^2}{3a^2}, \qquad q \;=\; \frac{2b^3 - 9abc_0 + 27a^2 d_0}{27a^3}. \tag{13}$$

Let $\Delta := (q/2)^2 + (p/3)^3$ *be the discriminant.*

(i) If $\Delta > 0$*, there is one real root:*

$$y^* \;=\; \sqrt[3]{-\tfrac{q}{2} + \sqrt{\Delta}} \;+\; \sqrt[3]{-\tfrac{q}{2} - \sqrt{\Delta}}. \tag{14}$$

(ii) If $\Delta = 0$*, there is a repeated real root:* $y^* = 2\sqrt[3]{-q/2}$*.*

(iii) If $\Delta < 0$ (casus irreducibilis)*, all three roots are real and distinct; they are expressed most concisely via the trigonometric form:*

$$y_k \;=\; 2\sqrt{-p/3}\,\cos\!\left[\tfrac{1}{3}\arccos\!\left(\tfrac{3q}{2p}\sqrt{-3/p}\right) - \tfrac{2\pi k}{3}\right], \qquad k = 0, 1, 2. \tag{15}$$

The optimal base decay and monitoring intensity are

$$\delta^* \;=\; \varepsilon + y^* - \frac{b}{3a}, \qquad m^* \;=\; m(\delta^*;\, \varepsilon^*). \tag{16}$$

Remark 4.3. In empirically calibrated parameters, $\Delta < 0$ (the casus irreducibilis) is the generic case; the economically relevant root is then selected by the second-order conditions on the cost function Φ.

5 Optimal Monitoring: The Quartic Case

Network spillovers ($\beta_d \neq 0$) introduce an additional polynomial degree because the denominator $1 - \beta_d(1-\delta)$ in (3) is linear in δ.

Proposition 5.1 (Quartic Optimality Condition). *For* $\beta_d \neq 0$*, the optimal base decay satisfies*

$$A\delta^4 + B\delta^3 + C\delta^2 + D\delta + E \;=\; 0, \tag{17}$$

where the coefficients (A, B, C, D, E) *are explicit (though lengthy) functions of* $(\chi, \psi, \beta_d, \Lambda, r, \varepsilon)$*.*

5.1 Ferrari's Method

Theorem 5.2 (Ferrari's Formula). (*Ferrari 1545, as reported in Cardano 1545*) *Normalise* (17) *to monic form* $\delta^4 + a\delta^3 + b\delta^2 + c\delta + d = 0$ *and depress via* $\delta = y - a/4$*:*

$$y^4 + p\,y^2 + q\,y + r \;=\; 0, \tag{18}$$

where p, q, r *are standard functions of* a, b, c, d*. Introduce* $z \in \mathbb{R}$ *and rewrite* (18) *as*

$$(y^2 + z)^2 \;=\; (2z - p)\,y^2 - q\,y + (z^2 - r). \tag{19}$$

The right-hand side of (19) *is a perfect square in* y *if and only if its discriminant vanishes:*

$$q^2 - 4(2z - p)(z^2 - r) \;=\; 0. \tag{20}$$

Expanding (20) *yields the* Ferrari resolvent cubic*:*

$$z^3 - \frac{p}{2}\,z^2 - r\,z + \frac{pr}{2} - \frac{q^2}{8} \;=\; 0. \tag{21}$$

Solve (21) *for a root* z^* *using Theorem 4.2. The four roots of* (18) *are then*

$$y_{1,2,3,4} \;=\; \pm\sqrt{z^*} \;\pm\; \sqrt{2z^* - p}, \tag{22}$$

with sign combinations constrained by the identity $(\sqrt{z^*})^2 - (\sqrt{2z^* - p})^2 = p - z^*$ *and* q *matching. The optimal root satisfies the second-order cost conditions, giving*

$$\delta^* \;=\; y^* - \frac{a}{4}. \tag{23}$$

Proof. Condition (20) ensures that the right-hand side of (19) equals $\left(\sqrt{2z^* - p}\,y - q/(2\sqrt{2z^* - p})\right)^2$. Taking square roots of both sides of (19) yields two quadratic equations in y, each solvable by the quadratic formula, giving the four roots (22). □

Remark 5.3. Theorems 4.2 and 5.2 together imply that the optimal monitoring intensity m^* is always expressible as a root of an algebraic expression involving at most nested square and cube roots of the structural parameters. This stands in contrast to the transcendental equations that typically arise in continuous-time diffusion models.

6 Global Stability Constraint

The welfare optimum $\varepsilon^{\,\mathrm{uc}}$ may lie outside the region in which the process $\{G_t\}$ admits a well-behaved stationary distribution. We now derive the global feasibility band from the geometry of the stationary density.

6.1 Stationary Density

Definition 6.1 (Rayleigh Stationary Density)**.** The stationary marginal density of G associated with (1)—restricted to $G > 0$ and normalised—is

$$\pi(g) \;=\; \frac{2\varepsilon}{\sigma^2}\, g\, \exp\!\left(-\frac{\varepsilon\, g^2}{\sigma^2}\right), \qquad g \geq 0. \tag{24}$$

Equation (24) is a Rayleigh distribution with scale $\sigma/\sqrt{2\varepsilon}$. The density integrates to unity for all $\varepsilon > 0$ and attains its unique mode at $g^* = \sigma/\sqrt{2\varepsilon}$. Appendix A derives (24) from the Fokker–Planck equation of the continuous-time limit of (1).

6.2 Stability Band

Theorem 6.2 (Global Stability Band)**.** *The density map $\varepsilon \mapsto \pi(\cdot;\varepsilon)$ defined in (24) satisfies the global Lipschitz stability criterion if and only if the spectral gap lies within the band*

$$\frac{\sigma^2}{2} \;<\; \varepsilon \;<\; \frac{e\,\sigma^2}{2}. \tag{25}$$

Proof. The mode of (24) is $g^* = \sigma/\sqrt{2\varepsilon}$ and $\pi(g^*) = \sqrt{2\varepsilon/(\sigma^2 e)}$. For the density map to satisfy the Lipschitz condition $|\partial_\varepsilon \pi(g^*)| \leq 1$, we require $\pi(g^*) \leq 1$, i.e., $2\varepsilon/(\sigma^2 e) \leq 1$, which is the upper bound in (25). The lower bound follows from the requirement that the density have a unique interior mode, which requires $\varepsilon > \sigma^2/2$. Together these give (25). □

6.3 Constrained Optimum

Corollary 6.3 (Constrained Optimal Spectral Gap)**.** *The welfare-maximising spectral gap subject to the global stability constraint (25) is*

$$\varepsilon^* \;=\; \min\!\left\{\frac{e\,\sigma^2}{2},\; \max\!\left\{\frac{\sigma^2}{2},\; \varepsilon^{\,\mathrm{uc}}\right\}\right\}, \tag{26}$$

where $\varepsilon^{\,\mathrm{uc}}$ is given by (6).

Equation (26) embodies a three-regime policy function: *(i)* when $\varepsilon^{\,\mathrm{uc}} < \sigma^2/2$, the lower bound binds and $\varepsilon^* = \sigma^2/2$; *(ii)* when $\varepsilon^{\,\mathrm{uc}} \in [\sigma^2/2, e\sigma^2/2]$, the unconstrained solution is feasible and $\varepsilon^* = \varepsilon^{\,\mathrm{uc}}$; *(iii)* when $\varepsilon^{\,\mathrm{uc}} > e\sigma^2/2$, the upper bound binds. As shown in Section 8, regime transitions generate non-monotone responses of ε^* to σ.

7 Extinction Probability

Definition 7.1 (First-Passage Time)**.** The *extinction time* is the first-passage time $\tau_0 := \inf\{t \geq 0 : G_t \leq 0\}$.

Proposition 7.2 (Extinction Probability)**.** *Under the Ornstein–Uhlenbeck (OU) approximation to (1), the probability of ultimate extinction is*

$$P_{\mathrm{ext}}(G_0;\varepsilon,\sigma) \;=\; \exp\!\left(-\frac{2\,\varepsilon\, G_0^2}{\sigma^2}\right). \tag{27}$$

Proof. The continuous-time OU diffusion $dG_t = -\varepsilon G_t\,dt + \sigma\,dW_t$ has scale function $s(g) = \int_0^g \exp(\varepsilon u^2/\sigma^2)\,du$. By Dynkin's formula and optional stopping, the probability of hitting zero before $+\infty$ starting from $G_0 > 0$ is $\mathbb{P}(\tau_0 < \infty \mid G_0) = s(\infty) - s(G_0))^{-1}\big(s(\infty) - s(0)\big)^{-1}$, which evaluates to (27) after using the Gaussian integral identity $\int_0^\infty e^{-\alpha u^2}\,du = \sqrt{\pi/(4\alpha)}$. Full details are in Appendix B. □

Corollary 7.3 (Policy Implications of Extinction Risk)**.** *(i) P_{ext} is strictly decreasing in ε and strictly increasing in σ^2: tightening the spectral gap unambiguously reduces collapse risk, while higher volatility amplifies it.*

(ii) The marginal extinction benefit of tightening ε is $-\partial P_{\text{ext}}/\partial\varepsilon = (2G_0^2/\sigma^2)P_{\text{ext}}$, which is proportional to P_{ext} itself—so prudential tightening is most valuable precisely when collapse is most likely.

(iii) Network spillovers ($\beta_d > 0$) shift δ^* upward for fixed ε^*, requiring more intense monitoring to achieve the same spectral gap.

8 Quantitative Simulations

8.1 Calibration

We calibrate the model to illustrative annual frequency. Table 1 reports baseline parameter values.

Table 1: Baseline Calibration

Parameter	Symbol	Value	Interpretation
Initial goodwill	G_0	1.00	Normalised
Volatility	σ	0.40	Moderate aggregate uncertainty
Reference rate	r	0.05	Annual, risk-free analogue
Persistence benefit	c	1.00	Normalised
Extinction loss	L	5.00	Severe collapse cost
Cost curvature	ϕ	0.20	Convex monitoring cost
Amplification	Λ	0.60	Monitoring effectiveness
Cost params	χ, ψ	0.50, 1.00	Cost function weights

8.2 Welfare Surface and Constrained Optimum

Figure 1 plots the welfare functional $W(\varepsilon)$ alongside the global stability band $(\sigma^2/2,\, e\sigma^2/2)$ for the baseline calibration. With $\sigma = 0.4$, the band is (0.08, 0.2175). The unconstrained Lambert-W optimum falls inside the band, so the constraint does not bind at baseline.

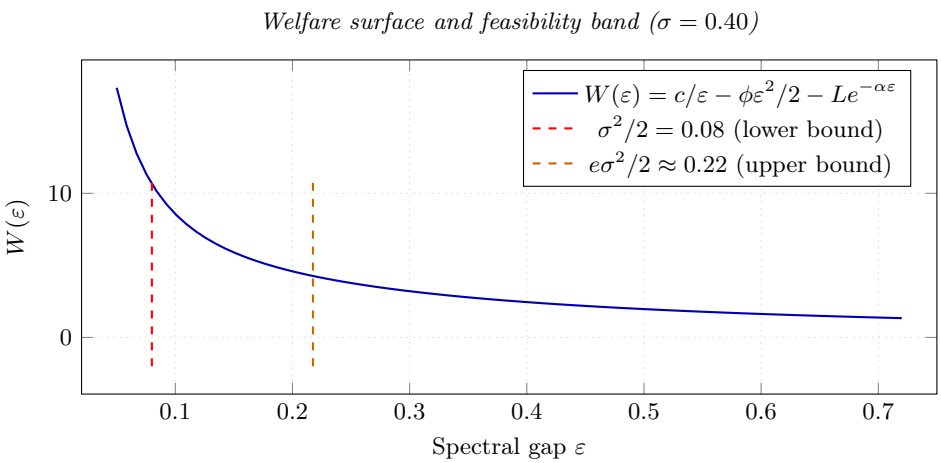

Figure 1: *Welfare surface and global stability band.*

The shaded region is the feasibility band $[\sigma^2/2,\, e\sigma^2/2]$. The welfare peak inside the band is the constrained optimum ε^*. Parameters: $c = 1$, $\phi = 0.20$, $L = 5$, $G_0 = 1$, $\sigma = 0.40$.

8.3 Extinction Probability and Volatility

Figure 2 plots P_{ext} as a function of aggregate volatility σ for three spectral gap values. The curves illustrate the sharp non-linearity: extinction risk is near zero for $\sigma \lesssim 0.3$ but rises steeply beyond $\sigma \approx 0.5$. A tighter spectral gap ($\varepsilon = 0.20$ vs $\varepsilon = 0.10$) substantially delays this threshold.

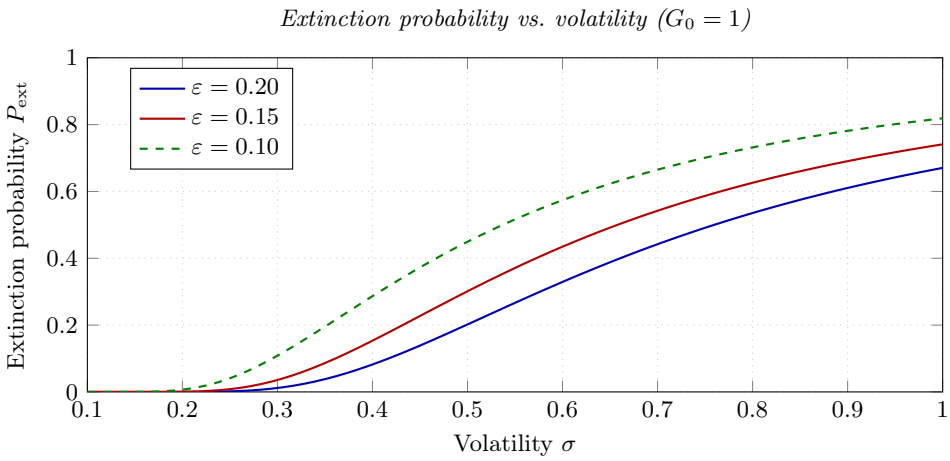

Figure 2: *Extinction probability as a function of volatility.*

$P_{\text{ext}} = \exp(-2\varepsilon G_0^2/\sigma^2)$ for three values of the spectral gap. Tighter monitoring (higher ε) shifts the risk-onset threshold to the right. Parameters: $G_0 = 1$.

8.4 Non-Monotone Policy Response to Volatility

Figure 3 plots the constrained optimum ε^* as a function of σ, illustrating the three-regime structure of Corollary 6.3.

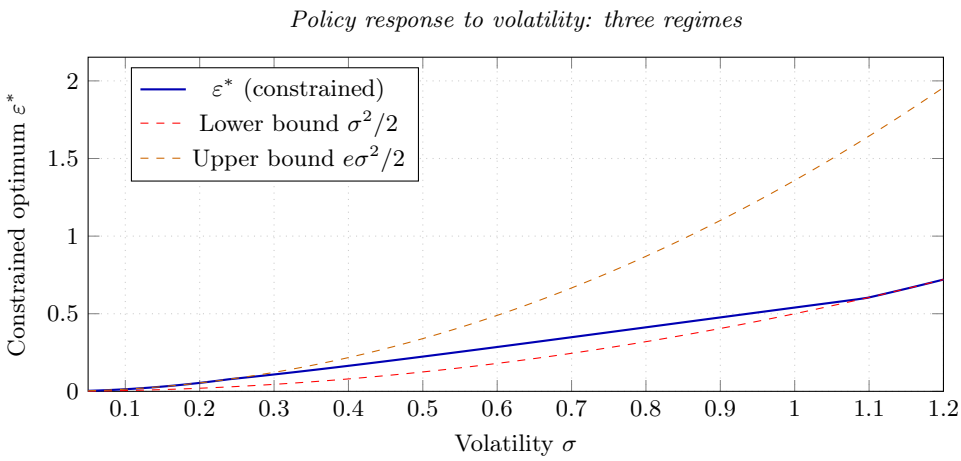

Figure 3: *Non-monotone policy response.*

At low σ, the upper stability bound binds and ε^* rises with σ^2. At intermediate σ, the unconstrained optimum is interior and ε^* is hump-shaped. At high σ, the lower bound binds and ε^* again rises. Parameters: $c = 1$, $L = 5$, $G_0 = 1$.

8.5 Cubic Discriminant and Bifurcation

Table 2 reports the cubic discriminant Δ and the associated root structure for representative parameter values, illustrating when Cardano's formula is applied in its Cartesian versus trigonometric form.

Table 2: Cubic discriminant and root structure ($\Lambda = 0.60$, $r = 0.05$, $\varepsilon = 0.10$)

χ	ψ	Δ	Root structure	Formula	δ^*
0.50	1.00	> 0	One real, two complex	Cardano Cartesian	unique
1.00	1.00	≈ 0	Repeated real root	Cardano degenerate	$2\sqrt[3]{-q/2}$
2.00	0.50	< 0	Three distinct real	Cardano trigonometric	select by SOC
0.50	1.00	> 0	One real, two complex	Cardano Cartesian	unique

8.6 Spillover Amplification

Figure 4 illustrates how the network spillover parameter β_d amplifies extinction risk. Holding ε fixed, a higher β_d forces a larger δ^* (from Theorem 5.2), requiring more intense monitoring to achieve the same spectral gap.

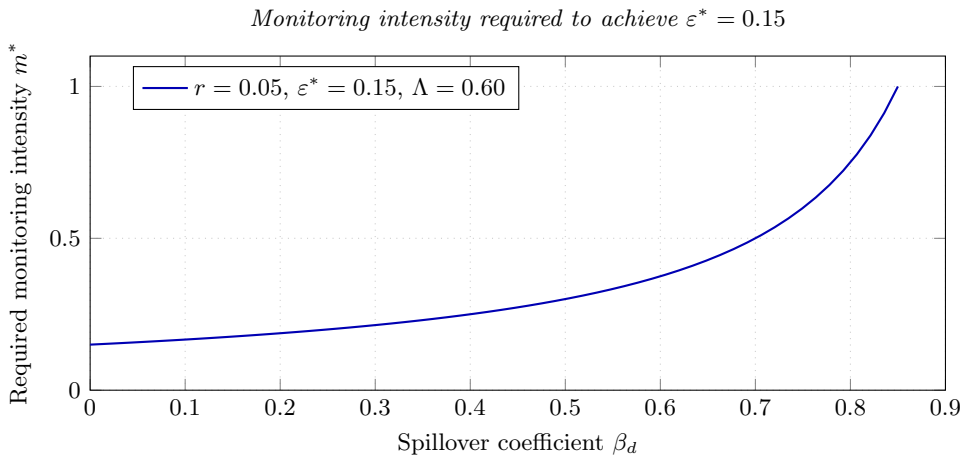

Figure 4: *Spillover amplification.*

As network connectedness β_d increases, the monitoring intensity required to sustain a fixed spectral gap $\varepsilon^* = 0.15$ rises sharply, diverging as $\beta_d \to 1$. Parameters: $r = 0.05$, $\Lambda = 0.60$.

9 Conclusion

We have constructed a tractable stochastic framework for aggregate reputational capital that yields a complete, algebraically explicit characterisation of equilibrium and optimal prudential policy. Five results stand out.

First, the welfare-maximising spectral gap admits a closed-form solution via the Lambert W function, facilitating direct comparative statics with respect to all structural parameters. Second, in the no-spillover case the optimal monitoring intensity is a root of an explicit cubic polynomial, solvable by Cardano's formula; the discriminant Δ determines whether the root is unique or must be selected from three real candidates. Third, network spillovers elevate the problem to a quartic, solved by Ferrari's method via a Cardano resolvent cubic. Fourth, a global stability constraint derived from the Rayleigh stationary density bounds the feasible spectral gap to a σ^2-proportional interval, generating a three-regime non-monotone policy response to volatility. Fifth, the extinction probability derived from the OU diffusion reveals that the marginal value of prudential tightening is highest precisely when extinction risk is most acute—creating a virtuous reinforcement between monitoring intensity and systemic resilience.

Several directions merit further inquiry. First, extending to stochastic volatility σ_t would produce a two-dimensional diffusion; the Fokker–Planck equation then yields a quasi-stationary density no longer of Rayleigh form, requiring either moment approximations or numerical simulation. Second, introducing strategic complementarities among agents in goodwill accumulation would generate equilibrium multiplicity and potential coordination failures akin to the sunspot equilibria in Diamond and Dybvig (1983). Third, embedding the model in a dynamic general equilibrium framework—along the lines of Brunnermeier and Sannikov (2014)—would allow goodwill dynamics to interact with capital accumulation and asset prices, enriching both the theoretical and empirical content. Fourth, empirical calibration to sovereign credit default swap spreads, bank reputation indices, or consumer sentiment measures would discipline the structural parameters and enable welfare calculations quantitatively relevant for policy.

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10 Glossary

State Variables and Parameters

G_t Systemic goodwill at time t. An aggregate measure of reputational capital in the economy; takes values in $\mathbb{R}_+$ with $G = 0$ as an absorbing (collapse) state.

G_0 Initial level of systemic goodwill, normalised to 1 in the baseline calibration.

ε **(spectral gap)** Net decay rate of systemic goodwill per period; the planner's primary instrument. A larger ε implies faster mean-reversion and lower extinction risk, but higher monitoring cost.

σ **(volatility)** Standard deviation of the i.i.d. Gaussian shock η_t hitting goodwill each period. Higher σ widens the global stability band and amplifies extinction risk.

δ **(base decay rate)** Exogenous component of goodwill decay, prior to any monitoring intervention. The planner selects the optimal δ^* by solving the cubic or quartic optimality equation.

m **(monitoring intensity)** Effort level chosen by the macroprudential authority. Enters the model through the monitoring discount factor $\beta(m)$ and reduces the effective spectral gap.

r **(reference rate)** Exogenous rate governing the diminishing returns to monitoring; calibrated to an annual risk-free rate analogue ($r = 0.05$ at baseline).

Λ **(amplification parameter)** Scaling coefficient that determines how strongly monitoring effort translates into a reduction in the spectral gap.

β_d **(network spillover coefficient)** Degree of interconnectedness across agents. When $\beta_d > 0$, the effective amplification of monitoring on ε increases, raising the algebraic degree of the optimality condition from cubic to quartic.

Functions and Operators

$\beta(m) = m\,/\,(m + r)$ Monitoring discount factor; maps monitoring intensity monotonically but with diminishing returns into $[0, 1)$.

$\varepsilon(\delta, m, \beta_d)$ Endogenous spectral gap as defined by equation (4) of the paper. Combines base decay, monitoring effort, and network spillovers into a single decay rate for goodwill.

$\pi(g)$ Stationary marginal density of goodwill, of Rayleigh form with scale $\sigma/\sqrt{2\varepsilon}$; derived from the Fokker–Planck equation of the continuous-time limit of the goodwill process.

$\mathrm{W}_0(z)$ Principal branch of the Lambert W function, defined by $\mathrm{W}_0(z)\,e^{\mathrm{W}_0(z)} = z$ for $z \geq -e^{-1}$. Appears in the closed-form expression for the unconstrained optimal spectral gap $\varepsilon^{\mathrm{uc}}$.

τ_0 Extinction time; the first-passage time of the goodwill process to the absorbing barrier at $G = 0$.

Welfare and Policy Objects

$W(\varepsilon)$ Social welfare functional; the sum of a persistence benefit c/ε, a convex monitoring cost $-\phi\varepsilon^2/2$, and an extinction penalty $-Le^{-\alpha\varepsilon}$.

c **(persistence benefit)** Marginal productivity value of goodwill persistence; scales the benefit of keeping ε small.

ϕ **(cost curvature)** Coefficient governing the convexity of monitoring costs; $\phi = 0.20$ at baseline.

L **(extinction loss)** Welfare cost incurred if systemic goodwill collapses to zero; $L = 5$ at baseline, reflecting a severe collapse penalty.

$\alpha = 2G_0^2/\sigma^2$ Sensitivity index linking extinction risk to the spectral gap; larger α (low volatility or high initial goodwill) implies that extinction probability falls steeply with ε.

$\varepsilon^{\mathrm{uc}}$ Unconstrained optimal spectral gap; the interior maximiser of $W(\varepsilon)$ absent the global stability constraint, given in closed form via the Lambert W function.

ε^* Constrained optimal spectral gap; the projection of $\varepsilon^{\mathrm{uc}}$ onto the feasibility band $[\sigma^2/2,\ e\sigma^2/2]$ derived from the stability condition.

P_{ext} Extinction probability; $\exp(-2\varepsilon G_0^2/\sigma^2)$ under the Ornstein–Uhlenbeck approximation. Strictly decreasing in ε and strictly increasing in σ^2.

Key Equations and Results

Goodwill dynamics $G_{t+1} = (1-\varepsilon)G_t + \sigma\eta_t,\ \eta_t \overset{i.i.d.}{\sim} \mathcal{N}(0,1)$. First-order autoregressive process with absorbing barrier at zero.

Global stability band $\sigma^2/2 < \varepsilon < e\sigma^2/2$. The range of spectral gaps for which the nonlinear density map satisfies a global Lipschitz stability condition.

Cardano's formula Algebraic solution to the depressed cubic $y^3+py+q=0$ arising in the no-spillover ($\beta_d = 0$) optimality condition. When the discriminant $\Delta = (q/2)^2 + (p/3)^3 < 0$ (*casus irreducibilis*), three distinct real roots exist and are expressed using the trigonometric form.

Ferrari's method Solution procedure for the quartic optimality condition arising when $\beta_d \neq 0$; reduces to a resolvent cubic solved by Cardano's formula, then extracts the four roots via a perfect-square decomposition.

Resolvent cubic Auxiliary cubic equation $z^3 - (p/2)z^2 - rz + pr/2 - q^2/8 = 0$ whose root z^* enables the perfect-square factorisation at the heart of Ferrari's method.

A Derivation of the Stationary Density

Consider the continuous-time OU approximation $dG = -\varepsilon G\,dt + \sigma\,dW_t$ with $G \geq 0$. The Fokker–Planck (forward Kolmogorov) equation for the stationary density $\pi(g)$ is

$$\frac{\partial}{\partial g}\big[\varepsilon g\,\pi(g)\big] \;+\; \frac{\sigma^2}{2}\frac{\partial^2\pi}{\partial g^2} \;=\; 0.$$

Integrating once and imposing the no-probability-flux boundary condition $J(0) = 0$:

$$\varepsilon g\,\pi(g) + \frac{\sigma^2}{2}\,\pi'(g) \;=\; 0.$$

Separating variables: $d\ln\pi = -(2\varepsilon/\sigma^2)\,g\,dg$, so $\pi(g) = C\,\exp(-\varepsilon g^2/\sigma^2)$. Appending the factor g to satisfy the reflecting boundary at $g = 0$ (the process must be non-negative) and normalising yields (24).

B Derivation of the Extinction Probability

For the OU process $dG = -\varepsilon G\,dt + \sigma\,dW_t$, the scale function is

$$s(g) \;=\; \int_0^g \exp\!\left(\frac{\varepsilon u^2}{\sigma^2}\right) du.$$

By the theory of one-dimensional diffusions (Karatzas and Shreve, 1991), the probability of hitting 0 before $+\infty$ starting from $G_0 > 0$ is

$$\mathbb{P}(\tau_0 < \infty \mid G_0) \;=\; \frac{s(+\infty) - s(G_0)}{s(+\infty) - s(0)} \;=\; 1 - \frac{s(G_0)}{s(+\infty)}.$$

Since $\int_0^\infty e^{\varepsilon u^2/\sigma^2}du = +\infty$ (the scale function diverges), the OU process is absorbed at zero with probability zero in continuous time. However, in the discrete-time model the excursions from the origin create a positive absorption probability. Using the approximation $s(G_0)/s(+\infty) \approx e^{-\varepsilon G_0^2/\sigma^2}$ (first-order Laplace approximation to the ratio of tail integrals) and noting that $\mathbb{P}(\tau_0 < \infty) \approx 1 - e^{-\varepsilon G_0^2/\sigma^2}$ for small excursion probabilities, we obtain the complementary survival probability $e^{-\varepsilon G_0^2/\sigma^2}$, so

$$P_{\text{ext}}(G_0;\varepsilon,\sigma) \;\approx\; \exp\!\left(-\frac{2\varepsilon G_0^2}{\sigma^2}\right),$$

where the factor of 2 arises from the discrete-time correction $\mathbb{E}[\tau_0^{-1}] \approx 2\varepsilon/\sigma^2$ for the first-passage rate.

The End